

## Thoughtlet

Thoughtlets are short articles that propose cross-fertilization of domains or promote new directions that might seed ideas for stimulating project management research. The editors of *Project Management Journal*<sup>®</sup> will invite distinguished subject matter experts to write the thoughtlets. The experts may come from other disciplines entirely or be long-time fixtures in the project management literature. We will ask these authors to express their ideas and show the potential to pursue these ideas. We will then publish these thoughtlets in a timely manner to stimulate timely dissemination within the research community.

# Factors versus Composites: Guidelines for Choosing the Right Structural Equation Modeling Method

Project Management Journal

Vol. 50(6) 619–624

© 2019 Project Management Institute, Inc.

Article reuse guidelines:

[sagepub.com/journals-permissions](https://sagepub.com/journals-permissions)

DOI: 10.1177/8756972819882132

[journals.sagepub.com/home/pmj](https://journals.sagepub.com/home/pmj)



Joseph F. Hair, Jr.<sup>1</sup> and Marko Sarstedt<sup>2,3</sup>

### Abstract

Structural equation modeling (SEM) is a widely applied and useful tool for project management scholars. In this Thoughtlet article, we critically reflect on the measurement philosophy underlying the two streams of SEM and their adequacy for estimating relationships among concepts commonly encountered in the field (e.g., team performance). We also discuss considerations to ponder when making the choice between the two types of SEM as well as between SEM and regression analysis.

### Keywords

composites, factors, partial least squares (PLS), structural equation modeling (SEM)

### Conceptual Variables, Factors, and Composites

In their quest to better understand and explain theory, researchers in project management and other fields of business research often deal with conceptual variables and theoretical models. A theoretical model reflects and visually represents a set of structural relationships—usually based on a set of equations connecting conceptual variables—that formalize a theory (Bollen, 2002). As elements of theoretical models, conceptual variables represent broad ideas or thoughts about abstract concepts that researchers establish and propose to measure in their research (e.g., team performance).

Structural equation modeling (SEM) is a family of statistical methods that estimate relationships among sets of observed

variables, which represent the conceptual variables in statistical models. There are two types of SEM methods, which differ in their assumptions about the way they map the elements of conceptual models in statistical analysis (Jöreskog & Wold,

<sup>1</sup> Mitchell College of Business, University of South Alabama, Mobile, AL, USA

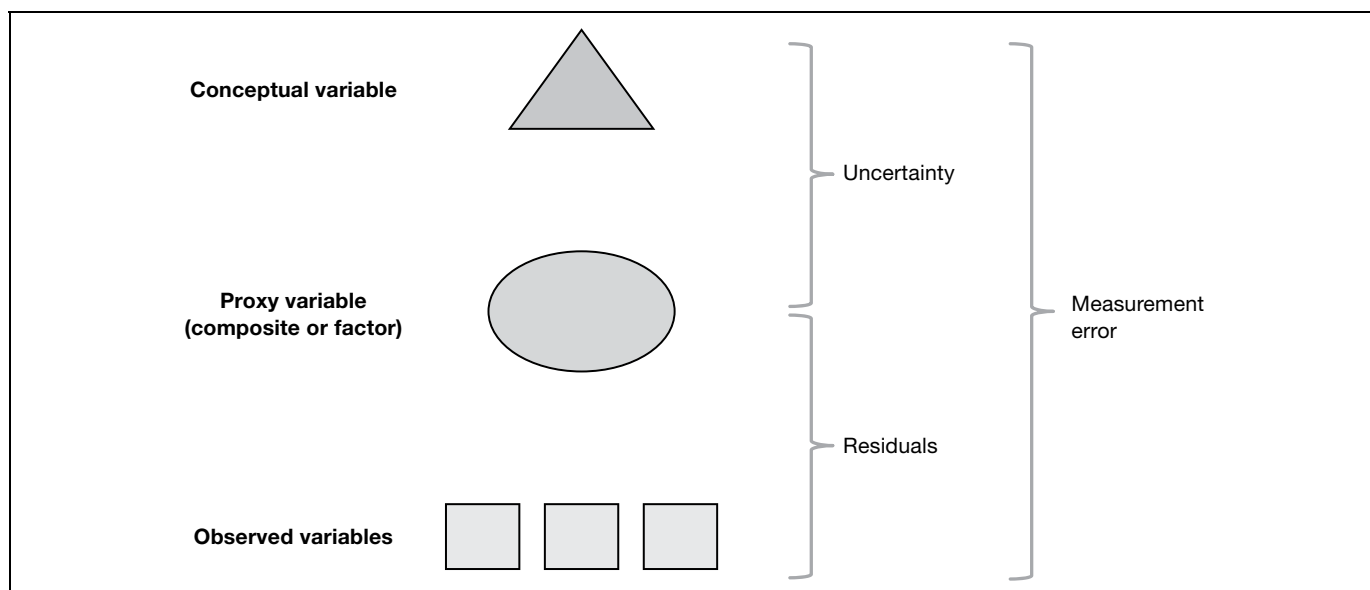
<sup>2</sup> Otto-von-Guericke University Magdeburg, Magdeburg, Germany

<sup>3</sup> School of Business and Global Asia in the 21st Century Research Platform, Monash University Malaysia, Selangor, Malaysia

### Corresponding Author:

Marko Sarstedt, Otto-von-Guericke University Magdeburg, Magdeburg, Germany; School of Business and Global Asia in the 21st Century Research Platform, Monash University Malaysia, Selangor, Malaysia.

Email: [marko.sarstedt@ovgu.de](mailto:marko.sarstedt@ovgu.de)



**Figure 1.** Measurement error framework.

1982). Factor-based SEM methods, as executed by software programs such as AMOS, Mplus, and LISREL, empirically represent the conceptual variable using common factors consisting of only common variance that explains the covariation between their associated indicators (Jöreskog, 1973). In contrast, composite-based SEM methods, such as partial least squares SEM (PLS-SEM; Wold, 1982) and generalized structural component analysis (GSCA; Hwang & Takane, 2004), use total variance to develop linear combinations of indicators to form composite variables that empirically represent the conceptual variables (Hwang, Sarstedt, Cheah, & Ringle, 2019; Khan, Sarstedt, Shiau, Hair, Ringle, & Fritze, 2019). Both SEM methods feature prominently in business research (e.g., Hair, Babin, & Krey, 2017; Hair, Hult, Ringle, Sarstedt, & Thiele, 2017) and have also gained traction in project management (Müller & Klein, 2018). For example, a search of *Project Management Journal*<sup>®</sup> for the 20-year period beginning 1999 through 2018 revealed 11 articles using factor-based SEM and 15 articles using composite-based SEM.

Regardless of researchers' preference for one method or the other, common factors and weighted composites serve as stand-ins for the unobservable conceptual variables in statistical models, and thus represent proxy variables (e.g., Teräsvirta, 1987; Wickens, 1972). The use of proxy variables is well established in many fields, such as in econometrics where researchers recognize their potential to create a confound (e.g., Frost, 1979). In contrast, the SEM literature and SEM users typically assume equivalence between the conceptual variable and the common factor or the weighted composite carrying the same name. If research is inspired by a theoretical model featuring the conceptual variable *team performance*, for example, researchers will refer to their factor or composite as *team performance*, and will discuss results involving this proxy as if

those results applied with precise equality to the conceptual variable (Tabassi, Roufechaei, Bakar, & Yusof, 2017).

A recent study by Rigdon, Becker, and Sarstedt (2019a), however, casts serious doubt on this common practice. These authors show that factor indeterminacy, which refers to the fact that the same observed variables are consistent with a multiplicity of different factor scores (i.e., the factor scores are mathematically indeterminate), induces a band of uncertainty in the relationship (i.e., the correlation) between a common factor inside the model and any variable outside the model (Steiger, 1979). Since the conceptual variable itself is not part of the model (i.e., it is outside the model), this range of correlation values applies as well to the relationship between an indeterminate common factor and the conceptual variable that it is designed to represent (Figure 1). For example, when measuring *team performance*, the presence of factor indeterminacy implies that the correlation between the common factor in the statistical model labeled *team performance* and the actual team performance is uncertain—even when the model exhibits perfect fit. For example, Rigdon et al. (2019a) found that in a commonly encountered model set-up with four moderately correlated (0.3) common factors, each measured with three indicators and all with 0.7 loadings, the correlation between a common factor (the construct labeled *team performance*) and the corresponding conceptual variable (the actual team performance) lies in a range whose width is 0.495. This means the range of uncertainty covers  $0.495/2 = 24.75\%$  of the total possible range of correlation (see Table 2 in Rigdon et al., 2019a).

Falsely equating common factors with conceptual variables suggests their preference over composites is unjustified. Some researchers argue that composite-based SEM methods are inferior to the factor-based SEM, in part because the latter approach accounts for measurement error (e.g., Rönkkö &

Evermann, 2013; Rönkkö, McIntosh, & Antonakis, 2015). As part of this argument, the observed variable residuals (i.e., the difference between the actual value of an observed variable and the predicted value) in the measurement models are identified as *measurement errors*. According to these scholars, the absence of residuals from composite-based approaches allegedly establishes that composite-based methods do not account for measurement error. Critics sometimes attempt to support this reasoning with simulation studies, which show that estimating factor model data using composite-based SEM methods produces biased parameter estimates—a finding not surprising considering that composite-based methods do not assume the data stem from a common factor model (Rigdon, Sarstedt, & Ringle, 2017; Sarstedt, Hair, Ringle, Thiele, & Gudergan, 2016).

More importantly, the argument that factor-based SEM accounts for measurement error, while composite-based methods do not, equates the observed variable residuals (see Figure 1) in a common factor model with measurement errors. However, observed variable residuals are only equal to measurement errors if the common factor is equivalent to the conceptual variable being measured. If that equivalence held, then the only discrepancy between observed variables and the conceptual variable would be the discrepancy between observed variables and the common factor (*residuals* in Figure 1). But if that equivalence does not hold, as Rigdon et al. (2019a) show (*uncertainty* in Figure 1), then there is no reason to assume observed variable residuals to be equivalent to measurement errors. In a simulation, of course, there is no conceptual variable—there is only the common factor in the simulation model—so questioning the equivalence of the conceptual variable and the proxy is meaningless.

In light of these results, the definition of measurement error should not be confined to the relationship between the observed variables and proxy variable, but should also consider the discrepancy between the conceptual variable and the proxy variable—as illustrated in Figure 1. This problem confronts researchers no matter what sort of proxy they choose—whether it is a common factor, a composite, or a single observed variable—and is problematic when asserting any strict preference for common factors as commonly observed in the social sciences and beyond (Rhemtulla, van Bork, & Borsboom, 2019). Or as Rigdon (2016, p. 602) notes, “common factor proxies cannot be assumed to carry greater significance than composite proxies in regard to the existence or nature of conceptual variables.”

## Implications for SEM Method Choice

Accepting that common factors may not be the gold standard to measure conceptual variables implies that the reflex-like preference for factor-based SEM methods prevalent in social science research is unjustified. Sarstedt et al. (2016) have demonstrated the pernicious consequences that can arise from estimating a composite model with factor-based instead of

composite-based SEM techniques. Their results show that (incorrectly) using PLS-SEM to estimate common factor models is much less of an issue than (incorrectly) using factor-based SEM to estimate composite models. The bias produced by factor-based SEM is, on average, 11 times higher than the bias produced by PLS-SEM when using each method on models inconsistent with what the methods assume (i.e., factor-based SEM on composite models and composite-based SEM on common factor models).

Since the underlying population model can seldom be known, composite-based PLS-SEM seems to be the safer choice when estimating relationships among conceptual variables. This is particularly true because PLS-SEM offers many advantageous features. For example, PLS-SEM enables researchers (1) to estimate complex models comprised of many latent and observed variables, and (2) to obtain solutions with small sample sizes without identification and convergence concerns (Hair, Risher, Sarstedt, & Ringle, 2019a). This characteristic is particularly beneficial in light of recent studies suggesting that researchers need to include more observed variables to measure proxies of conceptual variables—regardless of whether using common factor or composite proxies. More indicators reduce factor indeterminacy in common factors and increase the reliability of composite measures (Rigdon et al., 2019a; Sarstedt et al., 2016). However, since more indicators in a common factor model will likely lead to a worse  $\chi^2$  fit, researchers will have to assume more liberal thresholds in corresponding fit statistics, or abandon the use of fit statistics entirely—a bold call in light of researchers’ parental-like affection with model fit in factor-based SEM. At the same time, for composites, more observed variables could produce a high degree of collinearity, which in turn can lead to inflated standard errors and unstable parameter estimates. Researchers can avoid these undesirable side effects, however, by estimating correlation weights rather than regression weights (i.e., Mode A instead of Mode B in PLS-SEM; Rigdon, 2012).

As an alternative to PLS-SEM, researchers can employ the more recently developed GSCA approach, which has attracted considerable research attention among users of SEM and methodologists alike (Hwang et al., 2019). Like PLS-SEM, GSCA computes composites as stand-ins for the conceptual variables, but differs in model specification in that the method formulates a single equation to be optimized, which facilitates estimating all parameters in one stage. As a result, GSCA makes it possible to impose parameter constraints (Hwang & Takane, 2014). GSCA is a full-fledged approach to composite-based SEM, which has extended in many respects, for example, the ability to analyze functional data (i.e., data that arise from an underlying smooth function varying over a continuum such as in eye-tracker studies) or hierarchically structured data (Hwang, Takane, & Malhotra, 2007; Suk & Hwang, 2016).

To analyze their data, some researchers might be tempted to run a series of regressions using sum or mean scores of a set of observed variables measuring some conceptual variable as the input (Rönkkö et al., 2015). But the practice of using sum or

mean scores, also referred to as *parceling*, ignores the attenuating effect of measurement error that spreads through all path relationships and produces a combination of under- and over-estimation in the model estimates (Cole & Preacher, 2014). More importantly, Rigdon, Becker, and Sarstedt (2019b) have recently shown that ignoring the distorting effects of measurement error further increases the uncertainty in the relationship between the proxy variable and the conceptual variable (see Figure 1). As a consequence, the proxy variables potentially become further detached from the concepts that they represent (Steiger, 1979). Researchers should therefore always use SEM methods when dealing with unobserved conceptual variables as well as when estimating mediating effects (Hair, Sarstedt, & Ringle, 2019b). Despite contrary claims in the literature by Hayes, Montoya, and Rockwood (2017) that multiple regression is a better method than SEM—it is not.

## Conclusion and Outlook

This article offers a new perspective on the appropriateness of common factors and composites to represent conceptual variables and discusses the implications that arise for choosing the right SEM method. Table 1 summarizes our key arguments in the form of rules of thumb.

Recent research casting doubt on the universal appropriateness and applicability of the common factor model (e.g., Rhemtulla et al., 2019; Rigdon et al., 2019a, 2019b) has given further tailwind to composite-based SEM, which represent conceptual variables by weighted composites of observed variables. Corresponding methods such as PLS-SEM and GSCA offer researchers much flexibility in handling the progressively more complex interrelationships that populate today's theoretical models. For example, recent advances in composite-based SEM enable project management researchers to simultaneously analyze data from multiple levels of measurement. Potential areas of application include evaluating how company management impacts their project teams' inputs and outputs (Chiocchio & Hobbs, 2014) or analyzing the role of personality traits in project teams (Balthazard, Waldman, & Warren, 2009). Different from using standard regression-based hierarchical linear modeling with item parceling, composite-based SEM approaches allow accounting for measurement error in the model estimation.

Similarly, the ability to analyze functional data opens entirely new research avenues, such as in the implementation of virtual reality and augmented reality as a means of delivery and reinforcement of concepts and guidelines associated with existing and new work tasks (e.g., Ojiako, Ashleigh, Wang, & Chipulu, 2011). These recent technological advances could extend traditional approaches through real-time tracking and evaluation of the delivery of instructional images and messages onto what employees actually encounter in the workplace, providing immediate feedback on what is working and what needs improvement.

**Table 1.** Factors and Composites: Rules of Thumb.

- 
- Common factors and composites are both proxies for conceptual variables.
  - Factor indeterminacy elicits a validity gap between a common factor and the conceptual variable that it represents.
  - The preference for common factors over composites in social sciences research is not justified.
  - Common factor proxies and composite proxies both account for measurement error.
  - PLS-SEM and GSCA are both far superior to multiple regression with sum or mean scores of a set of observed variables—also when examining the possible influence of mediators and moderators.
- 

Finally, PLS-SEM and GSCA's emphasis on prediction in estimating statistical models whose structures are designed to provide causal explanations (Shmueli et al., 2019) makes them fit perfectly in today's research landscape. In order to fulfill the call for rigorous and relevant research, scholars not only need to test theories, but also offer managerial implications, which inherently follow a predictive paradigm (Hair et al., 2019a). Methods such as PLS-SEM and GSCA meet both these requirements. There are many applications in project management that call for such a causal-predictive approach to SEM. For example, composite-based SEM methods could be used to explore issues and obstacles associated with project management teams' adoption and implementation of blockchain technology (i.e., using software to create a distributed ledger that records transactions between two parties and is highly resistant to modification of the transaction data) in organizations. Another emerging area is the application of social robotics to improve efficiency and reduce costs. Social robotics involves using a robot instead of a person to interact and communicate with humans or other autonomous physical agents by following social behaviors and rules attached to its role. Again, project management scholars could apply PLS-SEM and GSCA to analyze data that explore issues and obstacles associated with adoption and implementation of social robotics to achieve organizational goals.

With their ongoing development, for example, in endogeneity (Hult, Hair, Proksch, Sarstedt, Pinkwart, & Ringle, 2018) or measurement invariance assessment (Henseler, Ringle, & Sarstedt, 2016), we expect the popularity of composite-based SEM to further increase, not only in the context of project management, which has readily adapted PLS-SEM and related methods (e.g., Jitpaiboon, Smith, & Gu, 2019; Müller, Turner, Andersen, Shao, & Kvalnes, 2016; Wen & Qiang, 2019), but in business and social sciences research generally.

## Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

## References

- Balthazard, P. A., Waldman, D. A., & Warren, J. E. (2009). Predictors of the emergence of transformational leadership in virtual decision teams. *Leadership Quarterly*, 20(5), 651–663.
- Bollen, K. A. (2002). Latent variables in psychology and the social sciences. *Annual Review of Psychology*, 53(1), 605–634.
- Chiocchio, F., & Hobbs, B. (2014). The difficult but necessary task of developing a specific project team research agenda. *Project Management Journal*, 45(6), 7–16.
- Cole, D. A., & Preacher, K. J. (2014). Manifest variable path analysis: Potentially serious and misleading consequences due to uncorrected measurement error. *Psychological Methods*, 19(2), 300–315.
- Frost, P. A. (1979). Proxy variables and specification bias. *The Review of Economics and Statistics*, 61(2), 323–325.
- Hair, J. F., Babin, B. J., & Krey, N. (2017). Covariance-based structural equation modeling in the *Journal of Advertising*: Review and recommendations. *Journal of Advertising*, 46(1), 163–177.
- Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., & Thiele, K. O. (2017). Mirror, mirror on the wall: A comparative evaluation of composite-based structural equation modeling methods. *Journal of the Academy of Marketing Science*, 45(5), 616–632.
- Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019a). When to use and how to report the results of PLS-SEM. *European Business Review*, 31(1), 2–24.
- Hair, J. F., Sarstedt, M., & Ringle, C. M. (2019b). Rethinking some of the rethinking of partial least squares. *European Journal of Marketing*, 53(4), 566–584.
- Hayes, A. F., Montoya, A. K., & Rockwood, N. J. (2017). The analysis of mechanisms and their contingencies: PROCESS versus structural equation modeling. *Australasian Marketing Journal*, 25(1), 76–81.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2016). Testing measurement invariance of composites using partial least squares. *International Marketing Review*, 33(3), 405–431.
- Hult, G. T. M., Hair, J. F., Proksch, D., Sarstedt, M., Pinkwart, A., & Ringle, C. M. (2018). Addressing endogeneity in marketing applications of partial least squares structural equation modeling. *Journal of International Marketing*, 26(3), 1–21.
- Hwang, H., Sarstedt, M., Cheah, J.-H., & Ringle, C. M. (2019). A concept analysis of methodological research on composite-based structural equation modeling: Bridging PLSPM and GSCA. *Behaviormetrika*, Advance online publication.
- Hwang, H., & Takane, Y. (2004). Generalized structured component analysis. *Psychometrika*, 69(1), 81–99.
- Hwang, H., & Takane, Y. (2014). *Generalized structured component analysis: A component-based approach to structural equation modeling*. Boca Raton, FL: Chapman and Hall/CRC.
- Hwang, H., Takane, Y., & Malhotra, N. (2007). Multilevel generalized structured component analysis. *Behaviormetrika*, 34(2), 95–109.
- Jitpaiboon, T., Smith, S. M., & Gu, Q. (2019). Critical success factors affecting project performance: An analysis of tools, practices, and managerial support. *Project Management Journal*, 50(3), 271–287.
- Jöreskog, K. G. (1973). A general method for estimating a linear structural equation system. In A. S. Goldberger & O. D. Duncan (Eds.), *Structural equation models in the social sciences* (pp. 255–284). New York, NY: Academic Press.
- Jöreskog, K. G., & Wold, H. O. A. (1982). The ML and PLS techniques for modeling with latent variables: Historical and comparative aspects. In H. O. A. Wold & K. G. Jöreskog (Eds.), *Systems under indirect observation, part I* (pp. 263–270). Amsterdam, the Netherlands: North-Holland.
- Khan, G., Sarstedt, M., Shiau, W.-L., Hair, J. F., Ringle, C. M., & Fritze, M. (2019). Methodological research on partial least squares structural equation modeling (PLS-SEM): A social network analysis. *Internet Research*, 29(3), 407–429.
- Müller, R., & Klein, G. (2018). What constitutes a contemporary contribution to *Project Management Journal*? *Project Management Journal*, 49(5), 3–4.
- Müller, R., Turner, J. R., Andersen, E. S., Shao, J., & Kvalnes, Ø. (2016). Governance and ethics in temporary organizations: The mediating role of corporate governance. *Project Management Journal*, 47(6), 7–23.
- Ojiako, U., Ashleigh, M., Wang, J.-K., & Chipulu, M. (2011). The criticality of transferable skills development and virtual learning environments used in the teaching of project management. *Project Management Journal*, 47(4), 76–86.
- Rhemtulla, M., van Bork, R., & Borsboom, D. (2019). Worse than measurement error: Consequences of inappropriate latent variable measurement models. *Psychological Methods*, Advance online publication.
- Rigdon, E. E. (2012). Rethinking partial least squares path modeling: In praise of simple methods. *Long Range Planning*, 45(5–6), 341–358.
- Rigdon, E. E. (2016). Choosing PLS path modeling as analytical method in European management research: A realist perspective. *European Management Journal*, 34(6), 598–605.
- Rigdon, E. E., Becker, J.-M., & Sarstedt, M. (2019a). Factor indeterminacy as metrological uncertainty: Implications for advancing psychological measurement. *Multivariate Behavioral Research*, 54(3), 429–443.
- Rigdon, E. E., Becker, J.-M., & Sarstedt, M. (2019b). Parceling cannot reduce factor indeterminacy in factor analysis: A research note. *Psychometrika* 84(3), 772–780.
- Rigdon, E. E., Sarstedt, M., & Ringle, C. M. (2017). On comparing results from CB-SEM and PLS-SEM: Five perspectives and five recommendations. *Marketing ZFP*, 39(3), 4–16.
- Rönkkö, M., & Evermann, J. (2013). A critical examination of common beliefs about partial least squares path modeling. *Organizational Research Methods*, 16(3), 425–448.
- Rönkkö, M., McIntosh, C. N., & Antonakis, J. (2015). On the adoption of partial least squares in psychological research: Caveat emptor. *Personality and Individual Differences*, 87, 76–84.
- Sarstedt, M., Hair, J. F., Ringle, C. M., Thiele, K. O., & Gudergan, S. P. (2016). Estimation issues with PLS and CBSEM: Where the bias lies! *Journal of Business Research*, 69(10), 3998–4010.
- Shmueli, G., Sarstedt, M., Hair, J. F., Cheah, J.-H., Ting, H., Vaithilingam, S., & Ringle, C. M. (2019). Predictive model assessment in

- PLS-SEM: Guidelines for using PLSpredict. *European Journal of Marketing*, 53(11), 2322–2347.
- Steiger, J. H. (1979). The relationship between external variables and common factors. *Psychometrika*, 44(1), 93–97.
- Suk, H. W., & Hwang, H. (2016). Functional generalized structured component analysis. *Psychometrika*, 81(4), 940–968.
- Tabassi, A. A., Roufehaei, K. M., Bakar, A. H. A., & Yusof, N. (2017). Linking team condition and team performance: A transformational leadership approach. *Project Management Journal*, 48(2), 22–38.
- Teräsvirta, T. (1987). Usefulness of proxy variables in linear models with stochastic regressors. *Journal of Econometrics*, 36(3), 377–382.
- Wickens, M. R. (1972). A note on the use of proxy variables. *Econometrica*, 40(4), 759–761.
- Wen, Q., & Qiang, M. (2019). Project managers' competences in managing project closing. *Project Management Journal*, 50(3), 361–375.
- Wold, H. O. A. (1982). Soft modeling: the basic design and some extensions. In K. G. Jöreskog & H. O. A. Wold (Eds.), *Systems under indirect observations, part II* (pp. 1–54). Amsterdam, the Netherlands: North-Holland.
- Multivariate Data Analysis*, Cengage Learning, UK, 8th edition, 2019 (cited 118,000+ times and one of the top five all time social sciences research methods textbooks); *Essentials of Business Research Methods*, Routledge, 4th edition, 2020; *Essentials of Marketing Research*, McGraw-Hill, 5th edition, 2020; and *A Primer on Partial Least Squares Structural Equation Modeling*, Sage, 2nd edition, 2017. He also has published numerous articles in scholarly journals such as the *Journal of Marketing Research*, *Journal of Academy of Marketing Science*, *Organizational Research Methods*, *Journal of Advertising Research*, *Journal of Business Research*, *Journal of Long Range Planning*, *Industrial Marketing Management*, *Journal of Retailing*, and others. He is writing a new book on *Marketing Analytics*, forthcoming in 2020 (McGraw-Hill). He can be contacted at [jhair@southalabama.edu](mailto:jhair@southalabama.edu)

**Marko Sarstedt** is a chaired professor of marketing at the Otto-von-Guericke-University Magdeburg (Germany) and Adjunct Professor at the Monash University Malaysia. His main research interest is the advancement of research methods to further the understanding of consumer behavior. His research has been published in, for example, *Journal of Marketing Research*, *Journal of the Academy of Marketing Science*, *International Journal of Research in Marketing*, *Organizational Research Methods*, *MIS Quarterly*, *Journal of the Association for Information Systems*, *Multivariate Behavioral Research*, and *Psychometrika*. Marko has co-edited several special issues of leading journals and co-authored four widely adopted textbooks, including “A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)” (together with Joe F. Hair, G. Tomas M. Hult, and Christian M. Ringle). He has been named member of Clarivate Analytic's Highly Cited Researcher List. He can be contacted at [marko.sarstedt@ovgu.de](mailto:marko.sarstedt@ovgu.de)

## Author Biographies

**Joseph F. Hair** is director of the PhD Program and Cleverdon Chair of Business in the Mitchell College of Business, the University of South Alabama, Mobile, Alabama, USA. Joe was recently recognized by Clarivate Analytics for being in the top 1% globally of all business and economics professors based on his citations and scholarly accomplishments, which for his career exceed 192,000. He has authored over 80 book editions, including *MKTG*, Cengage Learning, 13th edition, 2019;